Nearest Neighbors

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

One coherent neighbour-ranking record.

Trace distances and tie rules, compare k values and judge bounded validation evidence.

Keep one habit: W1–W3/W5 → selected three-point task → project method record → fresh changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Dataset Cards. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Label Quality. Give an example and a mistake that the idea helps you avoid.

Decision Trees

> **My remembered idea, example and correction**

Dataset Cards

> **My remembered idea, example and correction**

Label Quality

> **My remembered idea, example and correction**

W1 · Explain the mechanism

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main training A2 blocked,B4 clear,C8 clear,D10 blocked,E12 clear. Distance absolute difference; equal distances alphabetical ID; k=3 one vote each. Query3 distances1,1,5,7,9 → A/B/C clear; k=1 A blocked. Validation query/reference3blocked,9blocked,11clear gives nearest A/B/C,C/D/E,D/E/C and clear for all:1/3, same as training-majority clear baseline.

  Picture instead A2healthy,B8spotted,C11spotted,query5:3,3,6 and k=1healthy/k=3spotted; query7 gives5,1,4 and k=1spotted. Independent uses different A1healthy,B5spotted,C9healthy,query7.

- For query 3 cm, calculate **distance**s to all five **training** rows. Explain why A and B tie even though their labels differ.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Trace the current experiment

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main training A2 blocked,B4 clear,C8 clear,D10 blocked,E12 clear. Distance absolute difference; equal distances alphabetical ID; k=3 one vote each. Query3 distances1,1,5,7,9 → A/B/C clear; k=1 A blocked. Validation query/reference3blocked,9blocked,11clear gives nearest A/B/C,C/D/E,D/E/C and clear for all:1/3, same as training-majority clear baseline.

  Picture instead A2healthy,B8spotted,C11spotted,query5:3,3,6 and k=1healthy/k=3spotted; query7 gives5,1,4 and k=1spotted. Independent uses different A1healthy,B5spotted,C9healthy,query7.

- For query 3, list the selected IDs and **prediction**s for k=1 and k=3 using alphabetical ID order on equal **distance**s. Explain what changes and what remains the same.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Repair the reasoning

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main training A2 blocked,B4 clear,C8 clear,D10 blocked,E12 clear. Distance absolute difference; equal distances alphabetical ID; k=3 one vote each. Query3 distances1,1,5,7,9 → A/B/C clear; k=1 A blocked. Validation query/reference3blocked,9blocked,11clear gives nearest A/B/C,C/D/E,D/E/C and clear for all:1/3, same as training-majority clear baseline.

  Picture instead A2healthy,B8spotted,C11spotted,query5:3,3,6 and k=1healthy/k=3spotted; query7 gives5,1,4 and k=1spotted. Independent uses different A1healthy,B5spotted,C9healthy,query7.

- Reference for this exercise: open this week's student lesson, "Nearest Neighbors", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Why must the three validation reference labels not enter the **neighbour** pool? Give the fixed k=3 score and baseline score on the worked validation examples.

> **My reasoning and evidence**

MAIN LESSON · SECTION 16

W5 · Plan the next check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main training A2 blocked,B4 clear,C8 clear,D10 blocked,E12 clear. Distance absolute difference; equal distances alphabetical ID; k=3 one vote each. Query3 distances1,1,5,7,9 → A/B/C clear; k=1 A blocked. Validation query/reference3blocked,9blocked,11clear gives nearest A/B/C,C/D/E,D/E/C and clear for all:1/3, same as training-majority clear baseline.

  Picture instead A2healthy,B8spotted,C11spotted,query5:3,3,6 and k=1healthy/k=3spotted; query7 gives5,1,4 and k=1spotted. Independent uses different A1healthy,B5spotted,C9healthy,query7.

- Two of three **neighbour**s **vote** clear. Does that prove a two-thirds chance of a correct **prediction**? Does an exact feature match guarantee a correct label? Explain both limits.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected experiment task

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use only this selected task's A/B/C training points and query7. Rank by absolute length difference, then alphabetical ID on equal distance; select one or all three as requested. No reference label for the query is supplied, so report method predictions rather than accuracy or proof of health. Do not import the picture's point labels or the main five-row corridor pool.

- **Training** points: A length1 healthy, B length5 spotted, C length9 healthy. Query length7. Use absolute **distance**; for a nearest-**neighbor** tie choose the alphabetically earlier ID. Compare k=1 and k=3.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Changed independent case

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 9

Project · Keep the experiment evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Implement a transparent k-NN paper prototype for the Plant Classifier and record the chosen features, **scaling**, **distance**, and k.

- The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.
  - I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use W2: Apply **neighbour** and tie rules for Nearest **neighbour**s and **distance**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use the worked case for Nearest **neighbour**s and **distance**s. Proposed explanation: "A negative difference is a negative **distance**". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier learning target

REFERENCE · SECTION 5

Reference · Earlier learning target

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- **Training** lengths 2(H),6(S),10(H); query=8. Compute **distance**s. Explain the tie and why combining length in mm with width in cm changes a two-feature **distance**.

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 6

Extra practice · Earlier complete case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- One-dimensional **training** points: 2 healthy, 4 healthy, 8 spotted. Query=6; absolute **distance**.
- Find k=1 **neighbour**s and handle ties.
- Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · Changed case

REFERENCE · SECTION 7

Extra practice · Changed case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- For one feature, labeled points are 2:fern and 8:grass. A new point is 6. Under one-nearest-**neighbor** with absolute **distance**, predict its label.

- My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 12 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- **Training** lengths 2(H),6(S),10(H); query=8. Compute **distance**s. Explain the tie and why combining length in mm with width in cm changes a two-feature **distance**.

- Record support used: none, hint or teacher explanation.

My independently reasoned answer

▼ **Reference · Earlier W4 record**

REFERENCE · SECTION 15

Reference · Earlier W4 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Rules: use absolute difference in centimetres. Rank neighbours by distance, then alphabetical ID, and use the majority among the selected k training labels. Fit the constant baseline to the training-majority label; a majority tie would choose clear. Do not add validation rows to the neighbour pool.

- New **training** rows are P at 1 cm clear, Q at 5 blocked, R at 7 blocked, S at 11 clear, T at 13 blocked. For query 6, give all **distance**s, order the IDs by **distance** then alphabetically, and predict with k=1 and k=3.

  Using the same **training** pool and k=3, score validation (6,blocked),(10,clear),(12,blocked). Compare the **training**-majority baseline.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 18

Extra practice · Pictured case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

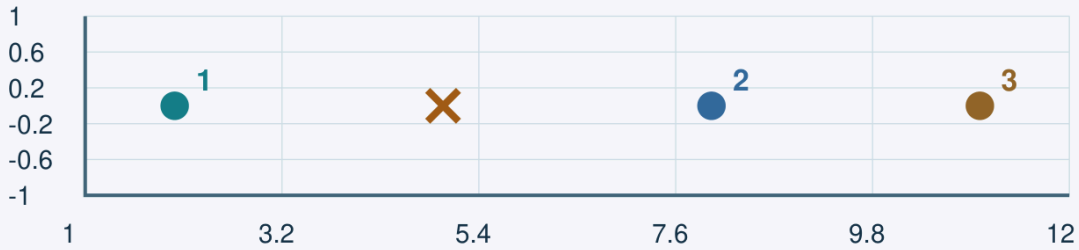
- Let a nearest-**neighbour** picture show each **distance** before voting.
- **Training**: A length 2 healthy; B length8 spotted; C length11 spotted. Query length5 cm. **Distance** is absolute difference; ties use alphabetically earlier ID.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- Why is the k=1 outcome at length5 not uniquely determined without a tie rule? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-4/12

One-dimensional neighbours shown along length axis



Horizontal: length (cm); vertical: fixed display row, y=0

1: A healthy at (2, 0)

2: B spotted at (8, 0)

3: C spotted at (11, 0)

Cross: query at (5, 0)

Distance rule: Absolute length difference; ties use alphabetically earlier ID

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/12

Nearest Neighbors

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Measure distances

To A: |5-2|=3; B: |5-8|=3; C: |5-11|=6 cm.

2. Apply k=1 rule

A and B tie nearest; declared ID tie-break chooses A, predicting healthy.

3. Compare k=3

All three vote: two spotted and one healthy, so k=3 predicts spotted.

CHANGE ONE THING

Move the query to length7 cm.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ Reference · Project portfolio milestone

REFERENCE · SECTION 19

Reference · Project portfolio milestone

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Project: Responsible Plant Classifier
- Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.
- For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 20

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Let a nearest-**neighbour** picture show each **distance** before voting.
- Which labelled step matches this detail? All three **vote**: two spotted and one healthy, so k=3 predicts spotted.
- 1. Measure **distance**s
- 2. Compare k=3
- 3. Apply k=1 rule
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Compare k=3 All three **vote**: two spotted and one healthy, so k=3 predicts spotted. Read the picture in order.

All three **vote**: two spotted and one healthy, so k=3 predicts spotted. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: Move the query to length7 cm. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. k=1 predicts spotted from B, **distance**1; k=3 remains spotted.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: k=1 predicts spotted from B, **distance**1; k=3 remains spotted. **Distance**s become 5, 1, 4 cm, so the nearest tie disappears.

Repair and keep the record

Review Decision Trees (ai-4/11). Goal: Fit a root question and leaf **prediction**s from **training** examples. Short practice: Which **input** distinguishes the original and changed tree?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Dataset Cards (ai-4/9). Goal: Inspect row meaning missing values and repeated records. Short practice: What two counts should the revised card show?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Label Quality (ai-4/6). Goal: Write an observable label rule with boundaries and a review option. Short practice: Why should two annotators not **vote** away Z's missing **evidence**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the relevant picture instead of repeating every task.

01. Nearest Neighbors: the case

AI ACADEMY / AI-4/12 · 01 · LOOK AT THE FACTS

Nearest Neighbors: the case

The situation

Let a nearest-neighbour picture show each distance before voting.

Supplied evidence

Training: A length 2 healthy; B length8 spotted; C length11 spotted. Query length5 cm. Distance is absolute difference; ties use alphabetically earlier ID.
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain nearest neighbors using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

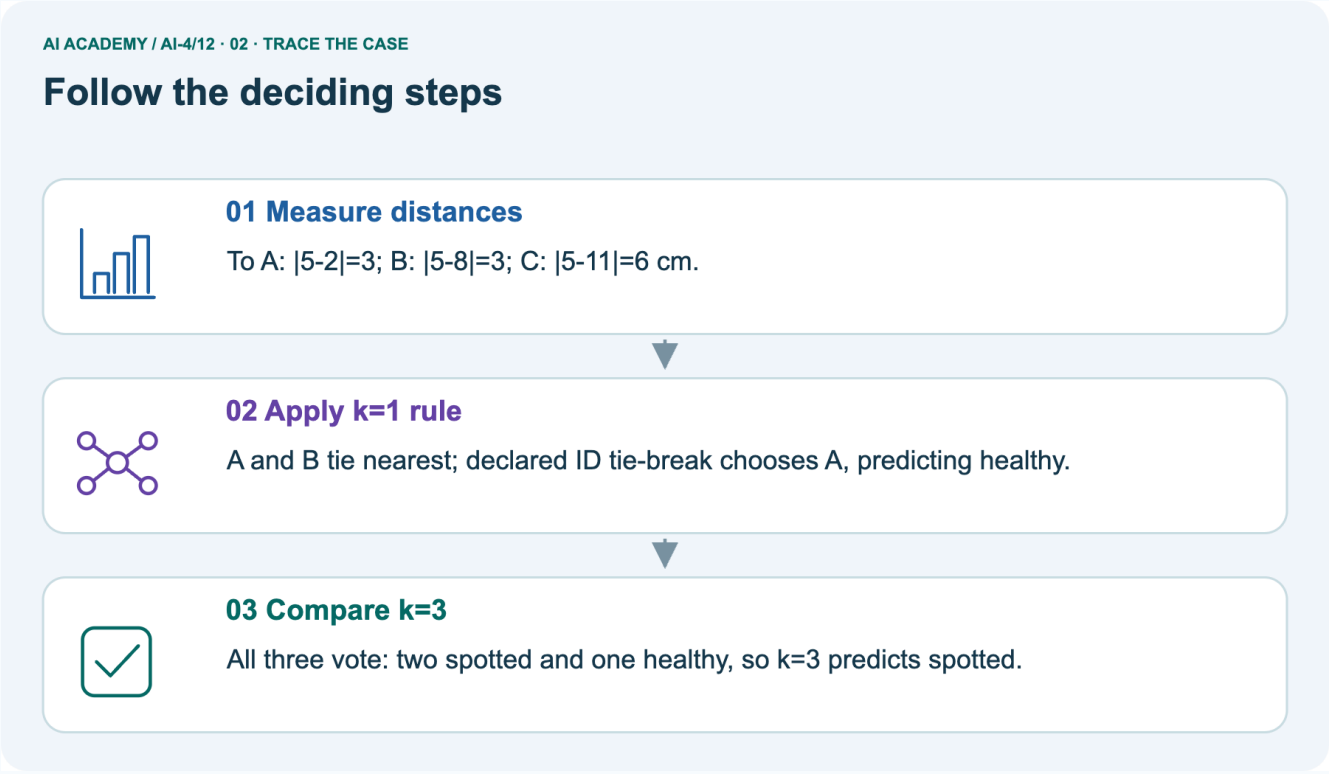
Let a nearest-**neighbour** picture show each **distance** before voting. Supplied facts: **Training**: A length 2 healthy; B length8 spotted; C length11 spotted. Query length5 cm.

Distance is absolute difference; ties use alphabetically earlier ID. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Measure **distance**s: To A: |5-2|=3; B: |5-8|=3; C: |5-11|=6 cm. → Apply k=1 rule: A and B tie nearest; declared ID tie-break chooses A, predicting healthy. → Compare k=3: All three **vote**: two spotted and one healthy, so k=3 predicts spotted.

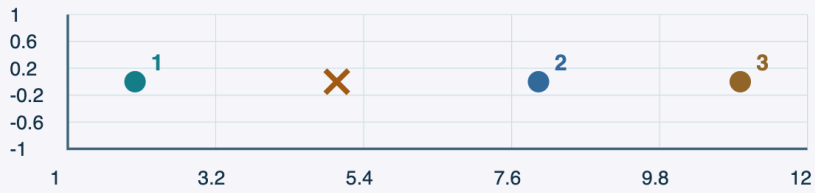
Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. One-dimensional neighbours shown along length axis

AI ACADEMY / AI-4/12

One-dimensional neighbours shown along length axis



Horizontal: length (cm); vertical: fixed display row, y=0

1: A healthy at (2, 0)

2: B spotted at (8, 0)

3: C spotted at (11, 0)

Cross: query at (5, 0)

Distance rule: Absolute length difference; ties use alphabetically earlier ID

Follow the labelled picture with the matching explanation. Open the vector diagram

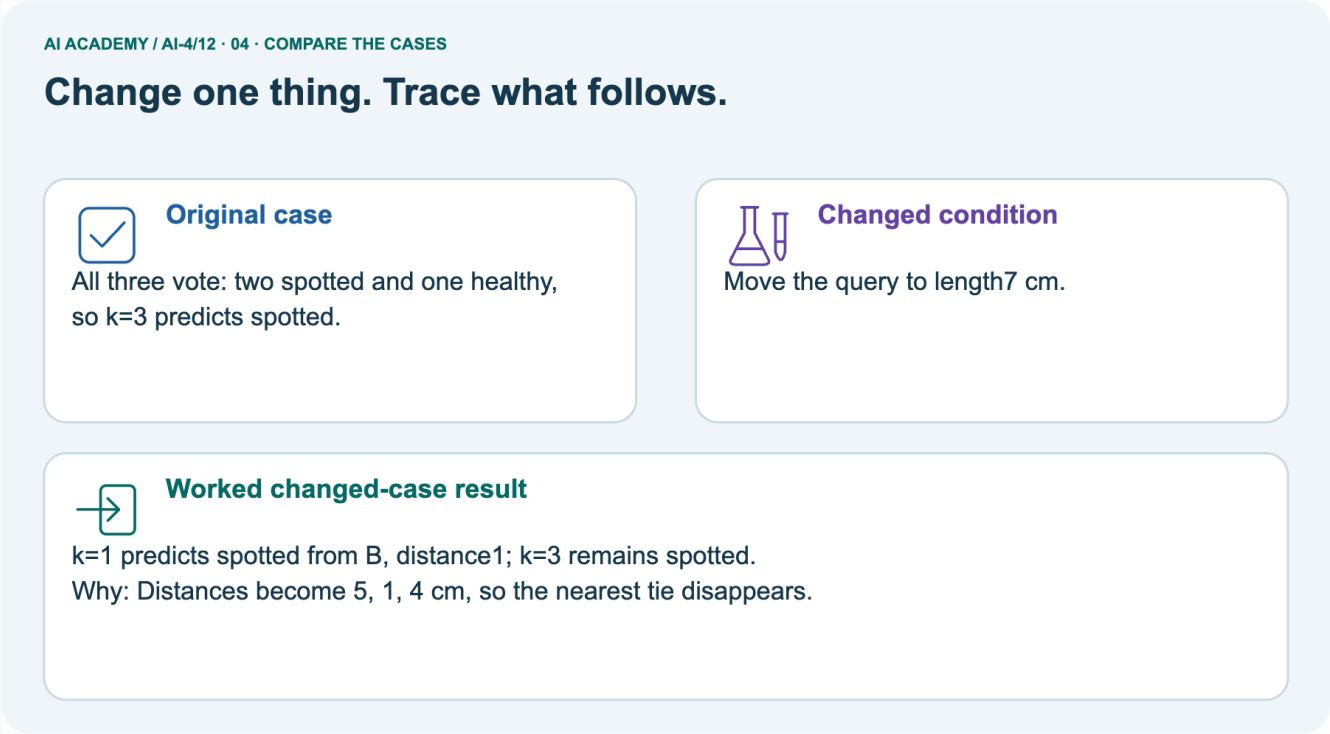
One-dimensional **neighbour**s shown along length axis. A healthy: (2, 0). B spotted: (8, 0). C spotted: (11, 0). Query query: (5, 0). Axes: length (cm); fixed display row, y=0.

Distance: Absolute length difference; ties use alphabetically earlier ID.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

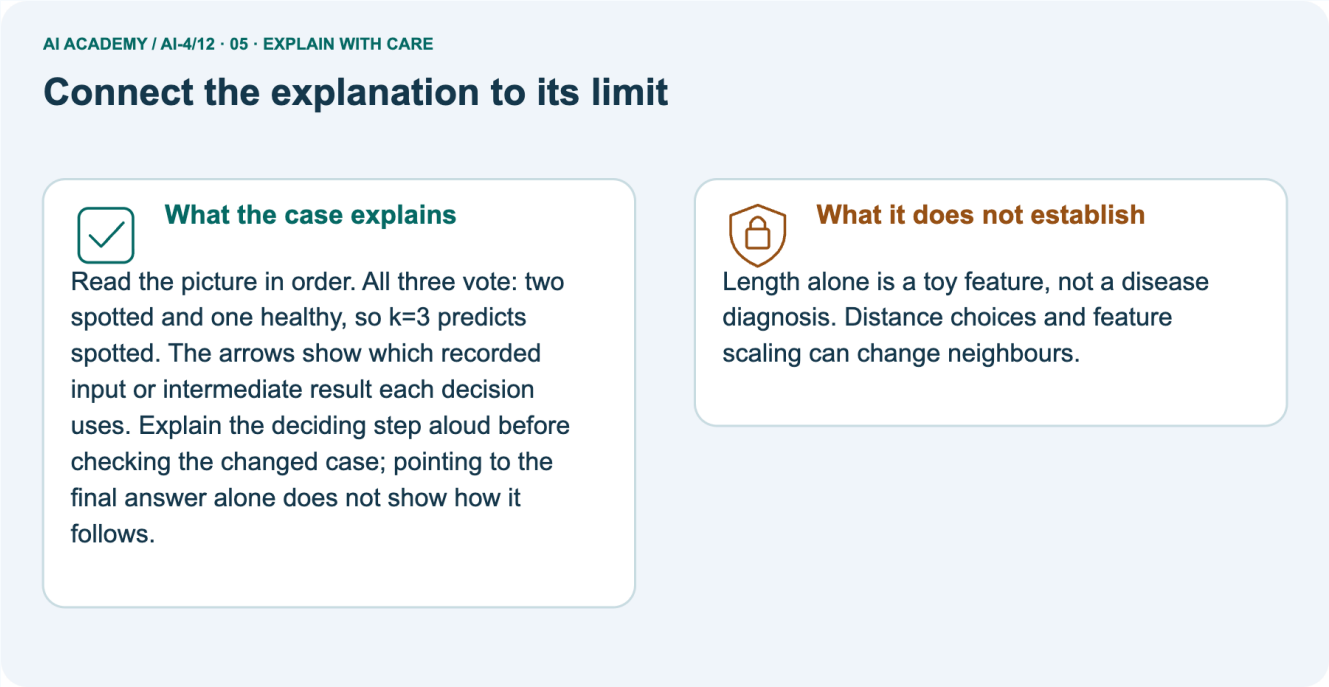
Original case: All three **vote**: two spotted and one healthy, so k=3 predicts spotted. Changed condition: Move the query to length7 cm. Expected result: k=1 predicts spotted from B, **distance**1; k=3 remains spotted.

Reason: **Distance**s become 5, 1, 4 cm, so the nearest tie disappears.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. All three **vote**: two spotted and one healthy, so k=3 predicts spotted. The arrows show which recorded **input** or intermediate result each decision uses.

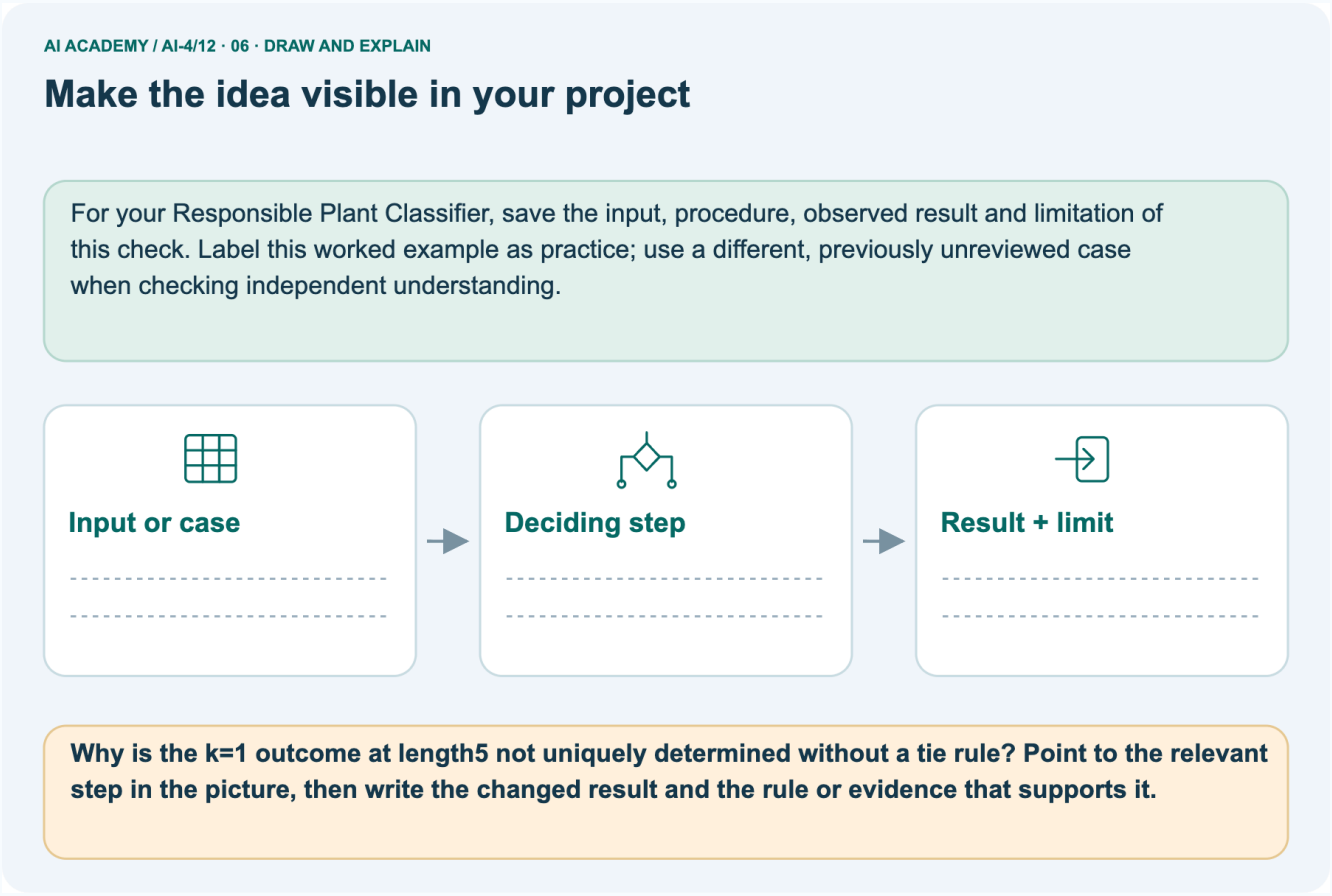
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Length alone is a toy feature, not a disease diagnosis.

Distance choices and feature **scaling** can change **neighbour**s.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why is the k=1 outcome at length5 not uniquely determined without a tie rule? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

01. Nearest Neighbors: the case

02. Follow the deciding steps

03. One-dimensional neighbours shown along length axis

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project
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<details>
<summary>Optional interactive practice and completed work</summary>

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.
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